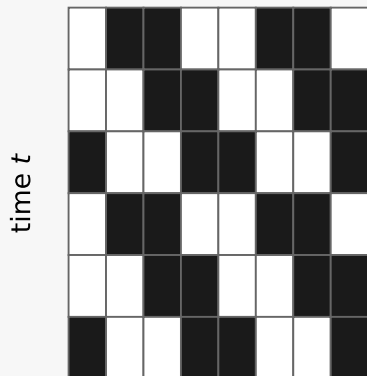


Relative-periodic domain selection by Bernoulli NML

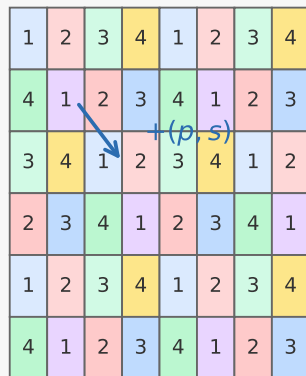
1 Observed spacetime

Enumerate candidate period/shift pairs (p, s) . The translation $(t, x) \mapsto (t + p, x + s)$ partitions the spacetime into orbit classes.



Example candidate: $p = 2$, shift $s = 1$.

2 Orbit classes



Each orbit shares one background bit.

3 Majority-vote background

Fit the nearest periodic background by a majority vote on each orbit class.

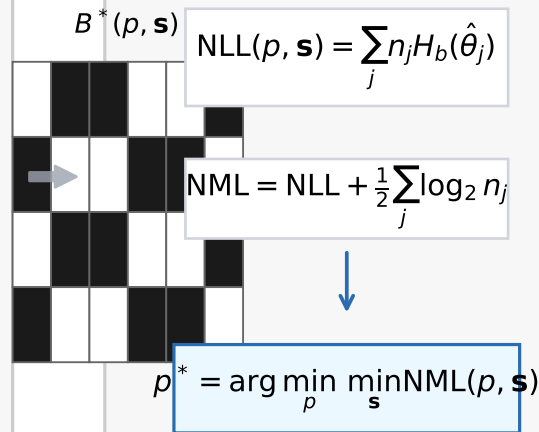
orbit	counts
1	5/1
2	1/5
3	4/2

ones / zeros

Residual mask: $M = U \oplus B^*$.

4 Score and select

Compute fit and complexity for every candidate, then keep the period with smallest score.



Output: periodic background + structured residual